

Cluster 15809

To view the latest Map of Science clusters and data, visit the live Map at sciencemap.eto.tech.

Subjects

Research fields: Engineering, Materials science, Environmental science

The most common research fields of articles in this cluster. [Read more.](#)

Research subfields: Petroleum engineering, Chemical engineering, Composite material

Common research subfields of articles in this cluster. [Read more.](#)

Key concepts: polymer flooding, enhanced oil recovery, Oil Polymer EOR, chemical enhanced oil, Heavy Oil

These key concepts are identified algorithmically using article titles and abstracts. [Read more.](#)

0.25% of all articles are AI-related

Articles are algorithmically labeled as AI-related or not. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to robotics

Articles are labeled as robotics-related or not using an ETO model. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to computer vision

Articles are labeled as computer vision-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to natural language processing

Articles are labeled as NLP-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to AI safety

Articles are labeled as related to AI safety or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to cybersecurity

Vital signs

804 recent articles in the cluster

804 articles in this cluster were published in the five year period ending in 2023.

Average article is **14.32** years old

The average age of articles in this cluster. However, most Map of Science metrics are calculated using articles from the past five years only.

Growth rating: **46.82th** percentile

Over the three year period ending in 2023, this cluster grew faster than roughly 46.82% of other clusters.

Citation rating: **56.05th** percentile

A cluster's citation rating is equal to the average citation percentile for all the articles in the cluster. Citation percentiles are calculated relative to articles of the same publication year. In this case, the average article in this cluster is cited more often than roughly 56.05% of all other articles published in the same year.

95.15% of articles are in English

Not expected to grow extremely fast

As determined by ETO's growth prediction model. [Read more.](#)

Countries and languages

Top author countries

Each row lists the number of articles in this cluster from the last five years with at least one author from an organization in the specified country. Data available for 708 of 804 articles. 502 of these articles had at least one author from an organization in these ten countries.

Country	Articles (five year period ending in 2023)	Yearly citations per article (average)
United States	142	2.24
China	137	1.72
United Arab Emirates	76	2.31
Saudi Arabia	49	2.31
Indonesia	36	0.62
India	31	2.00
Canada	30	2.61
Oman	28	1.06
United Kingdom	28	1.90
Brazil	27	1.66

Top funding countries

Data available for 165 of 804 articles. 142 of these articles disclosed funding from one or more organizations in these ten countries.

Country	Articles (last five years)	Yearly citations (average)
China	69	2.69
United States	18	3.91
Brazil	13	1.91
Saudi Arabia	12	1.82
Canada	11	4.01
United Arab Emirates	8	4.36
Colombia	6	1.62
Malaysia	4	2.65
Norway	4	1.46
Iran	3	1.20

Articles and sources

Core articles

Core articles are especially highly connected to other papers within the cluster.

[Polymer flooding in high temperature, high salinity conditions: Selection of polymer type and polymer chemistry, thermal stability](#)

2020: Journal of Petroleum Science and Engineering. 137 citations.

[Laboratory Evaluation of Polymer Retention in a Heavy Oil Sand for a Polymer Flooding Application on Alaska's North Slope](#)

2020: SPE Journal. 47 citations.

[Polymer flooding: Current status and future directions](#)

2023: Petroleum Science. 61 citations.

[Field Demonstration of the Impact of Fractures on Hydrolyzed Polyacrylamide Injectivity, Propagation, and Degradation](#)

2022: SPE Journal. 26 citations.

Extensional Effects during Viscoelastic Polymer Flooding: Understanding Unresolved Challenges

2020: SPE Journal. 40 citations.

[A comprehensive review of viscoelastic polymer flooding in sandstone and carbonate rocks](#)

2023: Scientific reports. 17 citations.

[Stability and Behavior in Carbonate Cores for New Enhanced-Oil-Recovery Polymers at Elevated Temperatures in Hard Saline Brines](#)

2020: SPE Reservoir Evaluation and Engineering. 63 citations.

Review articles

Review articles have between 100 and 1000 out-citations. These articles describe and systematize other contributors' research.

[Polymer flooding in high temperature, high salinity conditions: Selection of polymer type and polymer chemistry, thermal stability](#)

2020: Journal of Petroleum Science and Engineering. 137 citations.

[Advances in Understanding Polymer Retention in Reservoir Rocks: A Comprehensive Review](#)

2024: Polymer Reviews. 1 citations.

[Polymer Flooding: The Good, the Bad, and the Ugly - Lessons Learned from Field Practices, 2024, Day 3 Wed](#)

2024: 2024. 0 citations.

[An Analytical Tool to Predict Fracture Extension and Elastic Desaturation for Polymer Field Projects](#)

2023: SPE Annual Technical Conference and Exhibition. 1 citations.

[Review of Important Aspects and Performances of Polymer Flooding versus ASP Flooding](#)

2022: Bulletin of the Karaganda University. "Chemistry" series. 16 citations.

[A comprehensive review of viscoelastic polymer flooding in sandstone and carbonate rocks](#)

2023: Scientific reports. 17 citations.

[Recent Advancements in Viscoelastic Polymer Flooding EOR Applications in Carbonates and Sandstones, 2022, Day 3 Wed](#)

2022: 2022. 3 citations.

Highly cited articles

Highly cited articles are the articles in the cluster with the most citations overall.

[Polymer flooding in high temperature, high salinity conditions: Selection of polymer type and polymer chemistry, thermal stability](#)

2020: Journal of Petroleum Science and Engineering. 137 citations.

[Enhancing Heavy-Oil-Recovery Efficiency by Combining Low-Salinity-Water and Polymer Flooding](#)

2020: SPE Journal. 62 citations.

[Laboratory Evaluation of Polymer Retention in a Heavy Oil Sand for a Polymer Flooding Application on Alaska's North Slope](#)

2020: SPE Journal. 47 citations.

[Polymer flooding: Current status and future directions](#)

2023: Petroleum Science. 61 citations.

[Evaluation of Polymeric Materials for Chemical Enhanced Oil Recovery](#)

2020: Processes. 79 citations.

[Modeling of microflow during viscoelastic polymer flooding in heterogenous reservoirs of Daqing Oilfield](#)

2021: Journal of Petroleum Science and Engineering. 83 citations.

[Status and progress of worldwide EOR field applications](#)

2020: Journal of Petroleum Science and Engineering. 104 citations.

[Low Salinity Polymer Flooding: Effect on Polymer Rheology, Injectivity, Retention, and Oil Recovery Efficiency](#)

2020: Energy & Fuels. 67 citations.

[Stability and Behavior in Carbonate Cores for New Enhanced-Oil-Recovery Polymers at Elevated Temperatures in Hard Saline Brines](#)

2020: SPE Reservoir Evaluation and Engineering. 63 citations.

Top sources

The most common sources of articles from the last five years in this cluster, such as specific journals, conference proceedings, or preprint servers. Data available for 786 out of 804 articles. 187 of these articles came from these sources.

Source	Articles (last five years)
Polymers	36
Journal of Petroleum Science and Engineering	29
SPE Annual Technical Conference and Exhibition	24
SPE Journal	23
Day 3 Wed, November 17, 2021	14
SPE Improved Oil Recovery Conference	13
Day 2 Tue, September 01, 2020	12
SPE Reservoir Evaluation & Engineering	12
Journal of Petroleum Exploration and Production Technology	12
ACS omega	12

Authors, organizations, and funders

Top authors

The authors with the most articles from the last five years in this cluster. Data available for 799 of 804 articles. 112 of these articles were written by these authors.

Author	Articles (last five years)	Yearly citations (average)
Emad W. Al-Shalabi	45	4.55
Waleed AlAmeri	33	4.26
Kishore K. Mohanty	21	2.71
R. S. Seright	21	7.74
Anas M. Hassan	20	6.19
Shehadeh Masalmeh	20	2.78
Muhammad Shahzad Kamal	16	2.98
Shirish Patil	16	5.49
Madhar Sahib Azad	16	4.95
Guillaume Dupuis	15	2.97

Top author organizations

The author organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 750 of 804 articles. 220 had at least one author from one of these ten organizations.

Organization	Articles (last five years)	Yearly citations (average)
Khalifa University of Science and Technology - United Arab Emirates	51	2.39
The University of Texas at Austin - United States	40	2.2
King Fahd University of Petroleum and Minerals - Saudi Arabia	33	2.18
Research Institute of Petroleum Exploration and Development - China	26	3.27
China University of Petroleum, Beijing - China	22	2.55
Abu Dhabi National Oil (United Arab Emirates)	22	1.88
New Mexico Institute of Mining and Technology - United States	20	5.15
Daqing Oilfield General Hospital - China	20	2.09
China University of Petroleum, East China	19	1.55
Southwest Petroleum University - China	18	2.29

Organization types

This table covers the 666 articles in this cluster from the five year period ending in 2023 with data about their author organization types (out of 804 articles total). [Read more](#)

Commercial	Education	Nonprofit	Government
13.85%	35.88%	7.15%	1.44%

Top industry organizations

The industry organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 223 out of 804 articles. 131 of these articles had at least one author from an organization listed below.

Organization	Articles (last five years)	Yearly citations (average)
Abu Dhabi National Oil (United Arab Emirates)	22	1.88
Ecopetrol (Colombia)	18	1.26
China National Petroleum Corporation (China)	15	1.78
China National Offshore Oil Corporation (China)	15	1.73
Sinopec (China)	14	1.99
Saudi Aramco (United States)	14	2.51
Kuwait Petroleum Corporation (Kuwait)	14	0.92
Vedanta Resources (India)	12	1.17
Kemira (Finland)	11	1.95
Saudi Aramco (Saudi Arabia)	11	2.32

Top funders

Data available for 233 out of 804 recent articles. 93 of these articles disclosed funding from one or more of these organizations.

Funding organization	Articles (last five years)	Yearly citations (average)
National Natural Science Foundation of China	46	3.08
Khalifa University of Science & Technology	11	4.03
Ministry of Education - China	11	2.93
Ministry of Science and Technology - China	10	4.10
Ministry of Education - Brazil	9	1.70
Natural Sciences and Engineering Research Council - Canada	9	4.28
Khalifa University of Science and Technology - United Arab Emirates	7	4.98
United States Department of Energy - United States	7	5.11
National Council for Scientific and Technological Development - Brazil	7	2.08
China University of Petroleum, Beijing - China	7	3.12

Funder types

This table covers the 150 articles in this cluster from the last five years with data about their funding organization types (out of 804 articles total).

Commercial	Education	Nonprofit	Government
10.10%	27.40%	6.25%	56.25%

Patents and industry

158 patents cite articles in this cluster (**87.1th** percentile among all clusters)

6.36% of articles in the cluster are cited by patents

11.96% of articles have at least one author from industry

Top patent titles

Patent title	Same-cluster citations
ADDITION OF MONOVALENT SALTS FOR IMPROVED VISCOSITY OF POLYMER SOLUTIONS USED IN OIL RECOVERY APPLICATIONS	8
INJECTING MULTIPLE TRACER TAG FLUIDS INTO A WELLBORE	7
FLOW SYNTHESIS OF POLYMER NANOPARTICLES	7
POLYMERS AND NANOPARTICLES FOR FLOODING	7
MICROFLUIDIC CHIP WITH MIXED POROSITIES FOR RESERVOIR MODELING	7
DETERMINING OIL AND WATER PRODUCTION RATES IN MULTIPLE PRODUCTION ZONES FROM A SINGLE PRODUCTION WELL	7
MICROFLUIDIC CHIP WITH MULTIPLE POROSITY REGIONS FOR RESERVOIR MODELING	7
QUANTIFICATION OF POLYMER VISCOELASTIC EFFECTS ON SOR REDUCTION USING MODIFIED CAPILLARY	32
METHOD OF RECOVERING OIL EMBEDDED IN AN UNDERGROUND MATRIX ADJACENT A DEPLETED OIL RESERVOIR	17

Top patent assignees

Patent assignee	Same-cluster citations
UNIVERSITY OF ALBERTA	44
SAUDI ARABIAN OIL COMPANY	31
SATURN LICENSING, LLC	29
SCHLUMBERGER TECHNOLOGY CORPORATION	15
SCHLESWIG HOLSTEIN NETZ AG	15
IFP ÉNERGIES NOUVELLES	12
BASF (GERMANY)	11
SAUDI ARAMCO UPSTREAM TECHNOLOGY COMPANY	11
HAITHAM RAKAD MOHAMMAD AL HATAMLEH	10

Connections

Top citing clusters

Cluster	Number of significant citations	CSET extracted phrases
20337	129	enhanced oil recovery, Chemical Enhanced Oil, oil recovery mechanisms, nanoparticles, nanofluid
10366	125	enhanced oil recovery, chemical enhanced oil, surfactant adsorption, IFT, oil recovery factor
11981	106	enhanced oil recovery, Polymer Flooding, hydrophobically associating polymer, viscosity, hydrophobic
7605	87	polymer gel, preformed particle gel, enhanced oil recovery, gel system, particle gels
4448	65	low salinity water, oil recovery, salinity water flooding, wettability alteration, enhanced oil
50602	26	heavy oil recovery, heavy oil, enhanced heavy oil, crude oil, chemical flooding
30769	17	viscoelastic fluid flow, elastic turbulence, Weissenberg number, fluid flow, Couette flow
8822	15	Enhanced Oil Recovery, foam stability, foam flooding, Foam EOR, foam system
57031	15	enhanced oil recovery, EOR methods, EOR screening, EOR projects, oil production
16579	12	porous media, wettability, multiphase flow, capillary pressure, oil recovery

Top cited clusters

Cluster	Number of significant citations	CSET extracted phrases
10366	157	enhanced oil recovery, chemical enhanced oil, surfactant adsorption, IFT, oil recovery factor
11981	97	enhanced oil recovery, Polymer Flooding, hydrophobically associating polymer, viscosity, hydrophobic
4448	74	low salinity water, oil recovery, salinity water flooding, wettability alteration, enhanced oil
7605	49	polymer gel, preformed particle gel, enhanced oil recovery, gel system, particle gels
20337	42	enhanced oil recovery, Chemical Enhanced Oil, oil recovery mechanisms, nanoparticles, nanofluid
21612	25	polymer flooding, oil recovery, oil displacement, residual oil, Water Chemical Flooding
16579	18	porous media, wettability, multiphase flow, capillary pressure, oil recovery
57031	15	enhanced oil recovery, EOR methods, EOR screening, EOR projects, oil production
30769	15	viscoelastic fluid flow, elastic turbulence, Weissenberg number, fluid flow, Couette flow
15851	14	Xanthan gum, xanthan gum production, polysaccharide xanthan gum, rheological properties, gum solution

Top GitHub repositories

No data available.